UNIT 7 · STANDARD 6.EE.8

Graph Inequalities

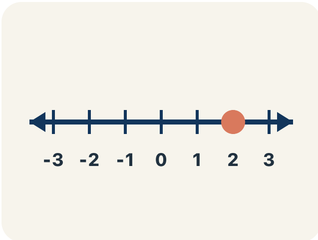
Lesson 7-5

Name: ______________ Date: ______________ Class: ______________

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



A horizontal line with marks at 0, 1, 2, 3, ... showing where values fall

Number line

Write the definition:

____

____

$x > 5$: open circle at 5 means 5 itself is NOT a solution, but 5.1, 6, 7, ... are

Open circle

Write the definition:

____

____

$x \geq 5$: closed circle at 5 means 5 IS a solution, and so are 6, 7, 8, ...

Closed circle

Write the definition:

____

____

For $x > 3$, the solution set includes 3.1, 4, 5, 10, 100 — any number greater than 3

Solution set

Write the definition:

____

____

x ≥ 3 is an inequality; it is graphed as a closed circle on 3 with an arrow to the right.

Inequality

Write the definition:

____

____

For x > 5, the boundary point is 5, shown with an open circle.

Boundary point

Write the definition:

____

____

Guided Notes

Level 2 Standard

🎯 **WHAT WE'RE LEARNING TODAY**

I can graph the solutions of an inequality on a number line.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Number line | **Open circle** | **Closed circle** | **Solution set** | **Inequality** | **Boundary point**

1. A line with numbers in order used to show values and solutions is a ____________ .

2. A hollow circle that shows a value is NOT included, used for < or >, is an ____________ .

3. A filled-in circle that shows a value IS included, used for ≤ or ≥, is a ____________ .

4. All the values that make an inequality true form the ____________ .

5. A math sentence comparing two amounts with <, >, ≤, or ≥ is an ____________ .

6. The number where the solution set begins on the number line is the ____________ .

🧮 Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 **I do — watch**

Follow each step as your teacher solves it.

Problem: Which graph represents x < 5?

A. Open circle at 5, shaded left

B. Closed circle at 5, shaded left

C. Open circle at 5, shaded right

D. Closed circle at 5, shaded right

Step 1 x < 5 means 5 is NOT included (open circle) and values less than 5 are shaded (left).

✅ **Answer:** A. Open circle at 5, shaded left

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: Which graph represents $x \geq 3$?

A. Closed circle at 3, shaded right

B. Open circle at 3, shaded right

C. Closed circle at 3, shaded left

D. Open circle at 3, shaded left

Step 1 ____________________

Step 2 ____________________

Answer: ____________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: Which graph represents x > 8?

A. Open circle at 8, shaded right

B. Closed circle at 8, shaded right

C. Open circle at 8, shaded left

D. Closed circle at 8, shaded left

Show your work:

____

____

____

Try It

Solve on your own. Check the answer key when you are done.

1. Clue: the witness ran more than 15 feet, so $d > 15$. On the number line, what do you draw?

A. Open circle at 15, shade right
B. Closed circle at 15, shade right
C. Open circle at 15, shade left
D. Closed circle at 15, shade left

Show your work:

2. Clue: the getaway car held at most 4 suspects, so $s \le 4$. How is this graphed?

A. Closed circle at 4, shade left
B. Open circle at 4, shade left
C. Closed circle at 4, shade right
D. Open circle at 4, shade right

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A theme park has two rides. Ride A requires height h > 48 inches. Ride B requires height h ≥ 48 inches. A child is exactly 48 inches tall. Which ride can they go on? Explain using number line graphs for both inequalities.

Sentence starter: For Ride A (h > 48), the graph has a ____ circle at 48. Since 48 > 48 is ____, the child ____ ride A. For Ride B (h ≥ 48), the graph has a ____ circle at 48. Since 48 ≥ 48 is ____, the child ____ ride B.

Show your work:

Reflect — Exit Ticket

How would you graph the inequality x ≥ 9?

A. Closed circle at 9, shaded right

B. Open circle at 9, shaded right

C. Closed circle at 9, shaded left

D. Open circle at 9, shaded left

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Number line
2. **Notes 2:** Open circle
3. **Notes 3:** Closed circle
4. **Notes 4:** Solution set
5. **Notes 5:** Inequality
6. **Notes 6:** Boundary point
7. **We Try (notes):** A. Closed circle at 3, shaded right *— $x \geq 3$ means 3 IS included (closed circle) and values greater than or equal to 3 are shaded (right).*
8. **You Try (notes):** A. Open circle at 8, shaded right *— $x > 8$ means 8 is NOT included (open circle) and values greater than 8 are shaded to the right.*
9. **Try It 1:** A. Open circle at 15, shade right *— 'Greater than' (>) is a strict symbol, so 15 is NOT included — open circle. Larger values work, so shade right.*
10. **Try It 2:** A. Closed circle at 4, shade left *— 'Less than or equal to' (≤) includes 4 — closed circle. Smaller values work, so shade left.*
11. **Exit Ticket:** A. Closed circle at 9, shaded right *— $x \geq 9$ means 9 is included (closed circle) and values are 9 or greater (shaded right).*